

Social Media World Model for Bystander Agent Training

1. Motivation: Task-Specific Social Media Agents

Constructing specialized social media agents for different scenarios is highly meaningful. For example, in cyberbullying settings, one could design a **bystander empowerment agent** that (1) comforts the victim and (2) actively counters the bully's harmful behavior. Such agents must operate within complex social dynamics involving multi-party interactions, reply chains, and public feedback mechanisms (e.g., likes, reposts, agreement/disagreement signals).

Recent works on LLM-based social simulation demonstrate that it is possible to model large-scale social interactions through agent-based frameworks [1]. Some works further integrate reinforcement learning (RL) into social media simulations to optimize an agent's influence or engagement within such environments [2]. These efforts suggest that simulated environments can serve as training grounds for social agents. However, existing approaches are not yet designed explicitly as **scalable world models for training generalizable policy agents across domains**.

2. Challenge: Delayed and Sparse Feedback in Real-World Social Media

A central difficulty in training social media agents lies in the nature of real-world feedback. Signals such as replies, upvotes/downvotes, reposts, and follower growth typically arrive over long time horizons. This creates two problems:

1. **Sparse and delayed rewards**, which make online RL inefficient.
2. **Limited opportunity for rapid iteration**, since real-world feedback cycles are slow and costly.

Although prior work has explored training RL agents within simulated social media environments [2], such training still depends on the simulation fidelity and computational tractability of the environment. In real platforms, policy optimization through direct online RL is impractical due to long feedback loops and safety concerns. Therefore, a **world model of social media** that can approximate global interaction dynamics becomes essential.

The survey of LLM-based social simulation highlights that current systems often focus on modeling social dynamics, evaluation of emergent behaviors, and validation of social theories, rather than providing dense, scalable training signals for RL policy learning [1]. This gap motivates the need for a new formulation of social media world models explicitly designed for RL-based agent training.

3. Limitations of Existing Social Media Simulation for Policy Training

There has been a surge of work on LLM-based social media simulation and agent-based modeling. However, directly using these systems for large-scale policy training faces two fundamental issues.

(1) Efficiency

Many recent systems rely on **LLM-based multi-agent simulations** to reproduce realistic interaction dynamics [1][3][4]. While these frameworks are effective for small-scale experiments and fine-grained analysis of social phenomena, they become computationally expensive when used as large-scale training environments.

For RL policy training, one typically requires:

- Massive numbers of rollouts,
- Fast environment steps,
- Stable and repeatable feedback signals.

Full LLM-based multi-agent interaction at every step introduces significant latency and cost, making it unsuitable as a high-throughput training signal generator.

(2) State / Signal Type Mismatch

Most existing social media simulation works are designed for **analyzing or validating social dynamics**, rather than for training policy agents.

Typical predicted states or targets include:

- Information propagation and diffusion patterns [3][4],

- Agent interaction structures and echo chambers [3],
- Emotional or opinion distribution dynamics [1],
- Public opinion distribution prediction [5].

These state types are valuable for understanding macro-level social phenomena. However, they differ from the signals required for RL-based policy training. A policy agent needs structured, actionable signals such as:

- Expected reply distributions,
- Engagement outcomes (e.g., like/repost counts),
- Counterfactual feedback under alternative actions.

Existing simulation works primarily aim at reproducing social theory or population-level statistics, rather than producing **policy-conditional transition dynamics** that support stable and scalable RL training.

4. Research Gap

In summary, although prior work has demonstrated:

- LLM-based social agent simulation at scale [1][3][4],
- Integration of RL into simulated social environments for influence optimization [2],
- Population-level opinion or engagement distribution prediction [5],

there remains a missing component:

A **computationally efficient social media world model** explicitly designed to provide training signals for RL-based policy agents, capturing global engagement dynamics (e.g., total replies, upvotes, repost distributions) in a form compatible with scalable policy optimization.

This proposal aims to bridge that gap.

References

[1] Wang et al. *A Survey on LLM-based Agents for Social Simulation: Taxonomy, Evaluation and Applications*. (Uploaded survey).

[2] Jin & Guo. *Build An Influential Bot In Social Media Simulations With Large Language Models*. arXiv:2411.19635.

[3] Zhang et al. *SocioVerse: A World Model for Social Simulation Powered by LLM Agents*. arXiv:2504.10157.

[4] Piao et al. *AgentSociety: Large-Scale Simulation of LLM-Driven Generative Agents*. arXiv:2502.08691.

[5] Mao & Tao. *MindVote: How LLMs Predict Human Decision-Making in Social Media Polls*. arXiv:2505.14422.